

# Homework 5

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## Functions and The Logistic Growth Model

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**1) Create a function that takes the parameters required to run a logistic growth model and returns a dataframe with columns for population size and time.**

The differential equation describing this process is below...

$$\frac{dN}{dt} = rN \left( 1 - \frac{N}{K} \right)$$

If you recall, the closed form solution to this differential equation requires an initial condition for population size,  $N_0$  and gives you a value for the population size at any given time,  $N(t)$ . This equation will be the one you code in your function.

$$N(t) = \frac{K}{1 + \left( \frac{K - N_0}{N_0} \right) e^{-rt}}$$

To get you started, I will list below the inputs your function will require to work properly.

- The initial population size,  $N_0$
- The growth rate,  $r$
- The carrying capacity,  $K$
- The end time for your time vector,  $t_{final}$
- The timestep for your time vector,  $t_{step}$

You should not need any global variables to do this. Keep all parameters as local within your function. The final step is to return your dataframe for plotting.

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## 2) Create a second function that takes the above dataframe as its input and creates a plot of population size over time as a side effect.

The hard part is over. Now you just need to create a second function that takes your dataframe and creates a plot. You can use ggplot or base graphics to do so. Remember to add meaningful axis labels. Now, you should be able to run your model and create a plot with different parameter sets by running only 2 lines of code.

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### Hints...

- Look up any mathematical operators we haven't covered explicitly in class. Ensure that parentheses and numerators/denominators are correctly positioned to capture the equation. This should be achievable with one line of code.
- Remember to refresh your function everytime you make a change or the function in memory will not include that change.
- Since we are not solving the differential equation directly/numerically (i.e. with a package like `deSolve`), we need to have a vector of times over which to evaluate this equation. Use your parameters to create your time vector within the function.
- Remember to add comments within your function listing the name, parameters, their required variable types/meaning, outputs, and overall purpose of the function.